Lagrangian and Eulerian Approach

The subject called **kinematics** concerns the study of *motion*. In fluid dynamics, *fluid kinematics* is the study of how fluids flow and how to describe fluid motion.

- A more common method of describing fluid flow is the **Eulerian description** of fluid motion.
- In the Eulerian description of fluid flow, a finite volume called a flow domain or **control volume** is defined, through which fluid flows in and out.
- Instead of tracking individual fluid particles, we define **field variables**, functions of space and time, within the control volume.
- The field variable at a particular location at a particular time is the value of the variable for whichever fluid particle happens to occupy that location at that time.
- For general unsteady three-dimensional fluid flow in Cartesian coordinates,

  Pressure field (scalar field variable): $\boldsymbol{P = P(x, y, z, t)}$

  Velocity field (vector field variable): $\boldsymbol{\vec{V} = \vec{V}(x, y, z, t)}$

  Acceleration field (vector field variable): $\boldsymbol{\vec{a} = \vec{a}(x, y, z, t)}$

- In the Eulerian description, all such field variables are defined at any location (*x, y, z*) in the control volume and at any instant in time *t.*

Lagrangian Approach:

- Plotting the position of an individual parcel through time gives the path line of the parcel.
- Key approach in Mechanics of Solids
- Properties of the element → **f (t)**, a function of time

$$\overline{R}_{\text{particle}} \equiv \text{Position vector}$$

$$\overline{V}_{\text{particle}} \equiv \frac{d}{dt}\overline{R}(t)$$

$$\overline{a}_{\text{particle}} \equiv \frac{d}{dt}\overline{V}(t)$$

Eulerian Approach:

- Observing a spatial point or region
- Has more mathematical implications
- Properties expressed as function (field function) of space and time

The difference between these two descriptions is made clearer by imagining a person standing beside a river, measuring its properties. In the Lagrangian approach, he throws in a probe that moves downstream with the water. In the Eulerian approach, he anchors the probe at a fixed location in the water.

Eulerian Approach: The airspeed probe mounted under the wing of an airplane measures the airspeed at that location.

The equations of motion in the Lagrangian description following individual fluid particles are well known (e.g. Newton's second law).

The equations of motion of fluid flow are not so readily apparent in the **Eulerian** description and must be carefully derived.

Lagrangian and **Eulerian derivatives** can be related through

1. Reynolds Transport Theorem (RTT)
2. Concept of Material derivative

Fluid kinematics is concerned with describing fluid motion, without necessarily analyzing the forces responsible for such motion. There are two fundamental descriptions of fluid motion—Lagrangian and Eulerian. In a Lagrangian description, we follow individual fluid particles or collections of fluid particles, while in the Eulerian description, we define a control volume through which fluid flows in and out. We transform equations of motion from Lagrangian to Eulerian through use of the **material derivative** for infinitesimal fluid particles and through use of the **Reynolds transport theorem (RTT)** for systems of finite volume.

Concept of Material derivative:

One dimensional:

$$\frac{Df}{Dt} = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial t}$$

Three dimensional:

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \vec{V} \cdot \nabla f$$

where

$$\vec{V} = u\hat{i} + v\hat{j} + w\hat{k};\ \nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

∂ is the partial derivative operator and d is the total derivative operator.

When we apply the material derivative to the velocity field, the result is the acceleration field, which is thus sometimes called the **material acceleration.**

$$\vec{a} = \vec{a}(x, y, z, t) = \frac{D\vec{V}}{Dt} = \frac{d\vec{V}}{dt} = \frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla)\vec{V}$$

$$\vec{a}(x, y, z, t) = \text{Material derivative} = \text{Total derivative} = \text{Local} + \text{Advective or Convective}$$

Material derivative of pressure:

$$P = \frac{DP}{Dt} = \frac{dP}{dt} = \frac{\partial P}{\partial t} + (\vec{V} \cdot \nabla)P$$

The above equation represents the time rate of change of pressure following a fluid particle as it moves through the flow and contains both local (unsteady) and advective components.

References:

1. Anderson, J. (2011)*: Fundamentals of Aerodynamics (SI units)*. McGraw hill.
2. Cengel, Y., & Cimbala, J. (2013)*: Fluid mechanics fundamentals and applications (SI units)*. McGraw Hill.